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Java scenario-based interview questions

virtusa

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🧪**Question** : Your endpoint works with @RequestMapping(method=POST), but not with @PostMapping. Why?

🧠 **Sample Answer:**

- @PostMapping introduced in Spring 4.3+ — older Spring versions don't recognize it

✅ **Fix:**

- Upgrade Spring
- Or use explicit @RequestMapping if version is locked

🧪**Question** : You expected computeIfAbsent() to run once, but it executes twice for same key. Why?

🧠 **Sample Answer:**
It's not atomic across threads if initial value is null and race occurs

✅ **Fix:**

- Surround with external sync if atomicity is critical
- Use putIfAbsent() with external value creation if heavy

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🧪**Question** : Your thread pool runs fine until a task throws an exception, then all tasks hang. Why?

🧠 **Sample Answer:**

- Task throws unhandled exception and crashes thread silently

✅ **Fix:**

- Use try-catch in tasks
- For ExecutorService, use submit() instead of execute() to capture exception in Future

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🧪**Question** : You configured retries, but RestTemplate doesn't retry on socket timeout. Why?

🧠 **Sample Answer:**

- Default RestTemplate doesn't support retry. You must configure with a custom ClientHttpRequestFactory.

✅ **Fix:**

- Use Apache HttpComponentsClientHttpRequestFactory + RetryTemplate
- Or switch to WebClient with reactive retries

🧪**Question** : Your code calls optional.isPresent() and then optional.get(), yet it still throws NoSuchElementException. Why?

🧠 **Sample Answer:**
You are calling get() on a different Optional — perhaps reassigned or shadowed

✅ **Fix:**

- Double-check variable shadowing
- Prefer:

```java
optional.ifPresent(value -> use(value));
```

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🧪**Question** : Actuator /health sometimes reports DOWN for your DB or service, even when everything looks fine. Why?

🧠 **Sample Answer:**

- Health indicator timeout or connection pool exhaustion

✅ **Fix:**

- Increase health check timeout
- Tune HikariCP max pool size or circuit breaker fallback

🧪**Question** : After a rebalance, your Spring Kafka consumer stops consuming messages. Why?

🧠 **Sample Answer:**
Listener container may have failed to reassign partitions or was stuck in error state

✅ **Fix:**

- Use ErrorHandler to handle exceptions
- Set:
- And monitor container lifecycle

```yaml
spring.kafka.listener.ack-mode: manual
```

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🧪**Question** : You set maxPoolSize = 10, but thread count goes to 50+. Why?

🧠 **Sample Answer:**

- You're using an unbounded queue, so new threads are created only if queue is full — and it never is

✅ **Fix:**

- Use a bounded queue (e.g., new ArrayBlockingQueue<>(100))
- Revisit pool + queue sizing strategy

🧪**Question** : You send a JSON payload, but Spring returns HTTP 400. Why?

🧠 **Sample Answer:**

- Common root cause: invalid field types or missing setters in DTO
- Jackson can't deserialize properly, but Spring doesn't throw detailed errors by default

✅ **Fix:**

- Check logs for deserialization stacktrace
- Use:

```java
@Validated
@ControllerAdvice
class GlobalExceptionHandler {
    @ExceptionHandler(MethodArgumentNotValidException.class)
```

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🧪**Question** : You try to fetch a custom header using @RequestHeader, but it's always null. Why?

🧠 **Sample Answer:**

- Either:
- Header not sent
- Case mismatch
- Proxy stripping headers

✅ Fix:

- Headers are case-insensitive, but proxies sometimes drop unknown headers
- Use tools like Postman or curl to debug

🧪**Question** : Your Spring REST client throws 415 Unsupported Media Type. Why?

🧠 **Sample Answer:**
Server expects Content-Type: application/json, but it's missing in client

✅ **Fix:**

- Check logs for deserialization stacktrace
- Use: Make sure to attach headers in request

```java
headers.setContentType(MediaType.APPLICATION_JSON);
```

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🧪**Question** : You loop over a collection and another thread modifies it — app crashes. Why?

🧠 **Sample Answer:**

- You're using non-thread-safe collection like ArrayList, HashMap

✅ Fix:

- Use CopyOnWriteArrayList or ConcurrentHashMap
- Or lock with Collections.synchronizedList(list)

🧪**Question** : Your scheduled job works under normal load but misses executions during heavy traffic. Why?

🧠 **Sample Answer:**
Default Spring scheduling is single-threaded — long-running tasks block the next run

✅ **Fix:**

```java
@EnableScheduling
@Bean
public TaskScheduler scheduler() {
  ThreadPoolTaskScheduler t = new ThreadPoolTaskScheduler();
  t.setPoolSize(5);
  return t;
}
```

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🧪**Question** : Your endpoint returns an object, but the response is always {}. Why?

🧠 **Sample Answer:**

- Cause: no public getters or field visibility is private + no getters
- Jackson fails to serialize but doesn't throw error

✅ Fix:

- Add proper getters or use @JsonProperty
- Alternatively, annotate with @Data (Lombok) or @JsonAutoDetect

🧪**Question** : Your REST client sends extra JSON fields, but they're silently dropped. Why?

🧠 **Sample Answer:**
The receiving class doesn't have those fields defined
✅ **Fix: Add field, or annotate with:** To safely ignore unknown fields

```java
@JsonIgnoreProperties(ignoreUnknown = true)
```

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